



Faculty of Mathematics and Computer Science

Introduction to Autonomous Driving Course (NLP)

Traffic Sign Recognition Using Semi-Supervised Learning with Virtual Adversarial Training

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Abstract

Traffic sign recognition is a fundamental component of **modern autonomous driving systems**, requiring accurate and reliable classification under a wide range of real-world conditions. Traditional Supervised Learning (SL) approaches have achieved strong performance in this domain but often depend on large amounts of labelled training data, which can be expensive and time-consuming to obtain. **Semi-Supervised Learning (SSL)** offers a practical alternative by combining a limited amount of labelled data with a larger pool of unlabelled samples.

This project investigates the application of **Virtual Adversarial Training (VAT)**, a **SSL** technique that improves model robustness by enforcing prediction consistency under small adversarial perturbations. Two neural network architectures were implemented and evaluated: a **Fully Connected Network (FCN)** and a **Small Convolutional Neural Network (SCNN)**. The models were tested in both artificial controlled environments and on the German Traffic Sign Recognition Benchmark (**GTSRB**) dataset under unsupervised (**UL**), semi-supervised (**SSL**) and fully supervised learning (**SL**) scenarios.

The experimental results show that **VAT** provides significant benefits when only a limited amount of labelled data is available. In the artificial experiments, **VAT** enabled meaningful learning even in fully unsupervised settings and generally accelerated convergence compared to traditional supervised training. As the proportion of labelled data increased, the performance gap between **VAT** and baseline approaches gradually decreased. The study demonstrates that semi-supervised learning (**SSL**) can effectively exploit unlabelled data and represents a promising direction for reducing annotation requirements in traffic sign recognition systems for autonomous driving applications.

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Keywords: (**AD**) Autonomous Driving; (**AI**) Artificial Intelligence; (**DL**) Deep Learning; (**FCN**) Fully Connected Network; (**GTSRB**) German Traffic Sign Recognition Benchmark; (**ML**) Machine Learning; (**ReLU**) Rectified Linear Unit; (**SCNN**) Small Convolutional Neural Network; (**SL**) Supervised Learning; (**SSL**) Semi-Supervised Learning; (**UL**) Unsupervised Learning; (**VAT**) Virtual Adversarial Training

1. Introduction

Machine Learning (ML) has become one of the most important areas of **Artificial Intelligence (AI)** because of its ability to learn patterns from data and make accurate predictions. Traditionally, **ML** methods are divided into two main categories: **Supervised Learning (SL)** and **Unsupervised Learning (UL)**.

In supervised learning **SL**, the model is trained using **labelled data**, meaning that each input example is associated with a correct output or target value. The objective is to learn a function that can correctly predict outputs for new and unseen data. Common applications of supervised learning include image classification, spam detection, medical diagnosis and speech recognition. On the other hand, unsupervised learning **UL** works with **unlabelled data**, where no output labels are provided. Instead of predicting a target value, the algorithm tries to discover hidden structures, patterns, or relationships inside the data. Clustering is one of the most common unsupervised learning tasks, where similar data points are grouped together based on their characteristics.

Semi-Supervised Learning (SSL) represents a hybrid approach that combines ideas from both supervised and unsupervised learning. In this learning paradigm, a small amount of labelled data is used together with a large amount of unlabelled data during the training process. The main goal is to improve learning performance by taking advantage of the additional information contained in unlabelled samples. This approach is particularly valuable because obtaining labelled data is often expensive, time-consuming or requires expert knowledge, while unlabelled data can usually be collected in large quantities with relatively low cost.

The importance of semi-supervised learning has increased significantly in recent years due to the rapid growth of data-driven applications. Many real-world domains generate huge amounts of raw data every day, but only a small portion of this data is manually annotated. For example, in medical imaging, doctors must carefully label scans and diagnostic data, which requires substantial expertise and effort. Similarly, in natural language processing tasks such as part-of-speech tagging or sentiment analysis, the annotation process can be labor-intensive and expensive. Semi-supervised learning **SSL** offers a practical solution by enabling models to learn useful information from both labelled and unlabelled datasets [14].

2. Literature Review

2.1. Semi-Supervised Learning

Semi-Supervised Learning (**SSL**) has become an important research area in machine learning **ML** because it combines the advantages of supervised learning **SL** and unsupervised learning **UL**. According to "van Engelen and Hoos" in paper "A survey on semi-supervised learning" [14], semi-supervised learning **SSL** aims to improve classification and prediction tasks by extracting useful information from unlabelled data.

In traditional supervised learning **SL**, the classifier only learns from labelled examples, which can limit its ability to generalize when the training data is small. **SSL** methods attempt to overcome this limitation by extracting additional structural information from unlabelled data.

"Van Engelen and Hoos" [14] explain that in supervised learning, the training dataset is represented as a collection of labelled samples:

$$D_L = ((x_i, y_i))_{i=1}^l$$

where x_i represents the input sample and y_i represents its corresponding label. The goal of supervised learning **SL** is to learn a function capable of predicting the label y^* for an unseen input x^* .

In semi-supervised learning **SSL**, an additional collection of unlabelled samples is introduced:

$$D_U = (x_i)_{i=l+1}^{l+u}$$

These unlabelled samples do not contain output labels, but they still provide useful information about the distribution of the data. **SSL** algorithms attempt to utilize both labelled and unlabelled datasets to construct models with better predictive performance than models trained using labelled data alone.

Another important contribution of **SSL** is its ability to improve decision boundaries. In classification problems, supervised learning **SL** algorithms may generate poor decision boundaries when only a few labelled samples are available. **SSL** uses the distribution of unlabelled data to better identify clusters and separate classes more effectively.

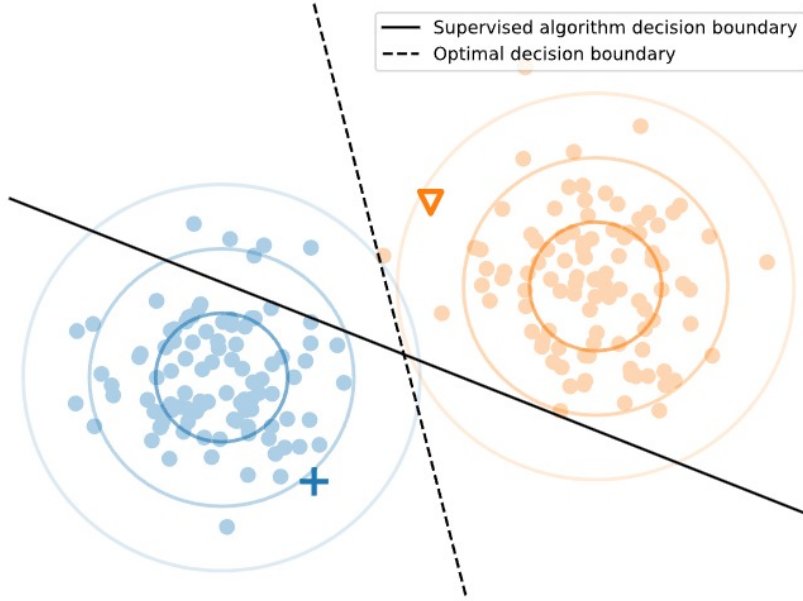


Fig. 1. Decision boundaries [14]

2.2. Semi-Supervised Learning Assumptions

Semi-supervised learning **SSL** relies on several theoretical assumptions that explain how unlabelled data can improve classification performance [14]. A necessary condition for **SSL** is that the marginal distribution $p(x)$ contains useful information about the posterior distribution $p(y|x)$. If the data distribution does not provide information about the labels, then unlabelled data cannot improve the learning process.

The main assumptions used in semi-supervised learning are summarized below:

- **Smoothness Assumption:** Data points that are close to each other in the input space are likely to share the same label. This allows SSL algorithms to propagate label information through neighbouring unlabelled samples;
- **Low-Density Assumption:** The decision boundary of a classifier should pass through low-density regions where few data points exist, rather than through dense clusters of samples. This helps reduce classification errors;
- **Manifold Assumption:** High-dimensional data often lies on lower-dimensional structures called manifolds. Data points located on the same manifold are expected to belong to the same class;
- **Cluster Assumption:** Samples belonging to the same cluster are likely to share the same label. This assumption connects closely with the smoothness, low-density, and manifold assumptions;

These assumptions form the theoretical foundation of most semi-supervised learning **SSL** algorithms and explain how unlabelled data can contribute to improved prediction performance.

2.3. Virtual Adversarial Training

Virtual Adversarial Training (VAT) was introduced by Miyato [9] as a **regularization method for supervised SL and semi-supervised learning SSL**. The main goal of VAT is to make a model's output stable when small changes are added to the input. This stability is called **Local Distributional Smoothness (LDS)**. In simple terms, a good model should not change its prediction too much when the input is slightly disturbed. VAT improves this smoothness by training the model to resist the most harmful small perturbation around each data point.

The important idea in VAT is that the perturbation is *virtual adversarial*. In adversarial training, the model requires the true class labels in order to find perturbations that can fool the classifier. However, VAT does not require label information, instead it measures the difference between the model's output distribution before and after a small perturbation is added to the input. The perturbation that maximizes this difference is selected as the virtual adversarial perturbation. Because labels are not required, VAT can be applied to unlabelled data, which makes it highly suitable for semi-supervised learning SSL.

Mathematically, VAT compares the prediction distribution at an input sample x with the prediction at a slightly perturbed sample $x + r$, where r is a small perturbation vector. The method searches for the perturbation direction that maximizes the **Kullback-Leibler (KL) divergence** between the two output distributions. The **KL divergence measures how different the two probability distributions are**.

The virtual adversarial perturbation is defined as [9]:

$$r_{adv} = \arg \max_{r: \|r\| \leq \epsilon} KL[p(y|x, \theta) \parallel p(y|x + r, \theta)]$$

where ϵ controls the size of the perturbation. After finding this perturbation, the model is trained to minimize the **KL divergence**. As a result, the model learns to produce similar outputs for both the original and perturbed inputs, creating a smoother decision boundary around the data points.

2.4. Virtual Adversarial Training in Semi-Supervised Learning

VAT is especially useful in semi-supervised learning SSL because **it can use both labeled and unlabelled data**. For labeled data, the model learns using the normal supervised classification loss, such as cross-entropy loss. For both labeled and unlabelled data, the VAT regularization term is added to encourage smooth predictions. This allows the model to learn the structure of the input distribution even when only a small amount of labeled data is available.

One challenge of VAT is that directly computing the worst perturbation direction is computationally expensive. To solve this problem, the authors [9] approximate the perturbation using the **Hessian matrix** and the power iteration method. In practice, only one iteration is usually enough to estimate the adversarial direction effectively. This makes **VAT computationally efficient and practical for deep neural networks**.

Its main contribution is improving the local smoothness of the model without requiring label information. Due to this property, VAT became an important method in semi-supervised learning SSL and influenced many later studies in **Deep Learning (DL)**.

2.5. Short Pseudocode of VAT

For each training step:

- Take a batch of labeled and unlabeled data
- Compute prediction $p(y|x)$

- Generate random direction d

- Estimate adversarial direction using gradient:

- $d = \text{gradient of KL divergence}$

- normalize d

```

Create perturbation:
    r_vadv = epsilon * d

Compute VAT loss:
    KL[p(y|x) || p(y|x+r_vadv)]

Update model using:
    supervised loss + VAT loss

```

3. Related Work

3.1. Problem Statement

Traffic sign recognition is an important component in **autonomous driving systems** [13]. Self-driving vehicles must correctly identify and interpret traffic signs in real time in order to make safe driving decisions. Incorrect recognition of traffic signs can lead to dangerous situations such as ignoring speed limits, entering restricted areas, or failing to stop at intersections.

The main problem addressed in this research is **the development of an accurate and computationally efficient traffic sign classification system** using modern deep learning DL methods. Although supervised learning SL approaches achieve strong performance, they usually require a large amount of labeled training data. In real-world autonomous driving scenarios, collecting and manually annotating large datasets is expensive and time-consuming. Because of this limitation, semi-supervised learning SSL has become an important research direction.

The experiments presented in this paper are conducted using the **German Traffic Sign Recognition Bench-mark (GTSRB)** dataset, a widely used benchmark in traffic sign recognition research and autonomous driving applications. The same dataset was also used in the reference paper “**Neural Network Based Traffic Sign Recognition for Autonomous Driving**” [13]. It is publicly available online at:

<https://www.kaggle.com/datasets/meowmeowmeowmeowmeow/gtsrb-german-traffic-sign>

The German Traffic Sign Benchmark was originally introduced as a multi-class, single-image classification challenge at the International Joint Conference on Neural Networks (IJCNN) 2011. It contains more than 50,000 traffic sign images distributed across more than 40 different traffic sign classes. Due to its diversity and large number of samples, the dataset has become one of the most commonly used benchmarks for evaluating deep learning DL models in traffic sign recognition tasks. The dataset includes realistic driving scenarios captured under various environmental and meteorological conditions. Images contain significant variations in illumination, viewing angle, scale, rotation, occlusion, background complexity, and image quality. These characteristics make the classification problem more challenging and closer to real-world autonomous driving conditions.

The semi-supervised learning SSL strategy aims to exploit both labeled and unlabelled samples during training. Pseudo-labelling techniques can be used to assign predicted labels to unlabelled images with high confidence scores. These pseudo-labeled samples are then incorporated into the training process, allowing the model to learn from a larger effective dataset. This approach helps improve robustness and classification accuracy while reducing the need for extensive manual annotation.

The final goal of this research is to evaluate whether a **Semi-Supervised Learning Virtual Adversarial Training-based (VATSSL)** model can be effectively implemented from scratch and achieve competitive or improved performance compared to widely used existing models.

3.2. Semi Supervised Learning Model with Virtual Adversarial Training

This project uses two different neural network architectures to recognize traffic signs: a **Fully Connected Network (FCN)** and a **Small Convolutional Neural Network (SCNN)**.

An FCN is a basic neural network where every node in one layer connects to every node in the next layer [1]. It processes images by flattening them into a single, one-dimensional array. For a 48 by 48 pixel image with 3 color chan-

nels, the starting size is 6,912 features. The FCN uses two hidden layers, each with 1,200 neurons, before outputting the final 43 classes representing the traffic signs.

The SCNN is designed specifically to process images. It uses three **Convolutional Blocks** [1]. A convolutional layer acts like a visual filter that scans small parts of the image to find patterns like edges and shapes. Each block contains two convolutional layers, followed by a **Max Pooling** layer. Max pooling shrinks the image data, keeping only the most important features. This makes the network faster and more efficient. Finally, a linear classifier layer maps the extracted features to the 43 output classes.

Both architectures rely on standard deep learning operations:

- **Linear and Convolutional Layers:** Perform the primary mathematical transformations by multiplying inputs with learned weights;
- **Batch Normalization:** Standardizes the data passing between layers. This makes the training process faster and more stable;
- **ReLU Activation:** The **Rectified Linear Unit (ReLU)** [10] is a simple mathematical function that helps the network learn complex patterns by only passing positive values forward;
- **Dropout:** A technique that randomly ignores a percentage of neurons during training. This forces the model to learn general patterns instead of simply memorizing the training data, a problem known as over-fitting.

3.3. Data Preprocessing

Before the neural network can learn from the images, the raw data must be prepared. This is called **data preprocessing**.

First, the original training data is split into two separate parts: **85% is used for training** and **15% is kept for validation**. A **validation set** is a group of images used to test the model's performance while it is still learning. This ensures the model is actually learning general patterns and not just memorizing.

Next, every image is resized to a standard size of 48 by 48 pixels. Neural networks require all input data to have the exact same shape to process them correctly.

To help the model perform well in unpredictable real-world conditions, a technique called **data augmentation** is applied to the training images [11]. Data augmentation artificially creates new training examples by slightly altering the original images. The model applies two main types of augmentations:

- **Geometric Changes:** Images are randomly rotated up to 15 degrees, shifted slightly from the center, zoomed, or flipped horizontally. This teaches the model to recognize a traffic sign even if the viewing angle is imperfect;
- **Color Changes:** The brightness, contrast, color saturation, and hue are randomly adjusted. This step simulates different environmental lighting conditions, such as driving at night or in bright sunlight.

After the images are resized and augmented, they are **normalized**. Normalization is a mathematical process that adjusts the pixel values so they fit within a standard scale [5]. This is done using the specific average colors of the German Traffic Sign dataset. Normalizing the data helps the neural network's math run much smoother.

Finally, the prepared images are grouped into **batches** of 128 images. A batch is a small chunk of the total dataset. Processing a batch of images at once is the most efficient way to use the computer's memory during training.

The **validation and testing images go through the same resizing and normalization steps**. However, **they are completely excluded from the random data augmentations**. This guarantees that the final evaluation of the model is fair, consistent and reproducible.

3.4. Model Training

The model training phase is where the neural network actually learns to recognize the traffic signs. The training process runs for a maximum of 20 **epochs** (due to limited time and resources). An epoch is one complete pass through all the training images.

To learn, the model must measure its own mistakes using a **Loss Function**, using **Cross-Entropy Loss** [8], which is a standard method for sorting images into categories. The loss function also applies **label smoothing** [12] rule,

forcing the network to be slightly less rigid and prevents it from becoming overly confident in incorrect answers. Because this is a semi-supervised project, this standard loss is only calculated for images that have known labels.

To learn from unlabeled images, the model uses **VAT**.

After calculating the mistake, an **Optimizer** adjusts the model's internal math so it performs better on the next image. This project uses an optimizer called **AdamW** [7]. AdamW includes a safety feature called **weight decay**. Weight decay keeps the internal numbers of the model small, stopping the model from simply memorizing the training data.

The size of the adjustment the optimizer makes is called the **learning rate**. A **Scheduler** is used to manage the learning rate automatically, using the **Cosine Annealing** [6] method. It starts with a higher learning rate so the model learns big, obvious patterns quickly, then smoothly lowers the rate over the 20 epochs. This allows the model to carefully fine-tune its details at the very end.

The training process follows a repeated cycle during each epoch:

1. **Training Phase:** The model processes batches of images and updates its parameters based on the prediction errors and **VAT**;
2. **Validation Phase:** The model is evaluated on clean, non-augmented validation images without updating its parameters. This monitors performance and checks for over-fitting;
3. **Checkpoint Saving:** If the model achieves a higher validation accuracy, the current model parameters are saved as a **checkpoint**.

To save time, a rule called **early stopping** is applied. If the model does not improve its validation accuracy for 15 epochs in a row, the training process stops early. After training is complete, the best saved checkpoint is loaded and evaluated on a completely unseen test dataset. This gives the final, objective measurement of the model's performance.

4. Experiments

To evaluate the effectiveness of the proposed approaches, the project is divided into two main experimental stages: **artificial experiments** and **real-world experiments using the GTSRB dataset**.

The purpose of the artificial experiments is to study the behaviour of different learning methods in a controlled environment. These experiments use synthetically generated datasets "Circles and Moons", where the data distribution and class structure can be easily visualized and analysed. This allows the strengths and limitations of each learning strategy to be observed before applying them to a more complex image classification problem.

The second stage focuses on real-world experiments using the **GTSRB**. This dataset contains images of 43 traffic sign categories collected under realistic driving conditions. Compared to the artificial datasets, it presents a more challenging classification task due to variations in lighting, viewing angle, occlusion, image quality and background content.

The project implements several neural network models based on different training strategies. Each model is evaluated under multiple supervision settings in order to measure how the amount of available labelled data affects performance.

The experiments are organized into the following learning scenarios:

- **Unsupervised Learning UL:** The model is trained without using class labels. In this setting, the objective is to learn useful feature representations directly from the input data;
- **Semi Supervised Learning SSL:** Only a portion of the training samples are associated with labels, while the remaining samples are treated as unlabelled data. Semi-supervised techniques such as Virtual Adversarial Training are used to exploit both labelled and unlabelled information;
- **Supervised Learning SL:** All training samples are provided with labels. This represents the traditional supervised learning setting and serves as a reference point for comparison with the other approaches.

For both the artificial datasets and the **GTSRB** dataset, the same experimental protocol is applied. Each model is trained and evaluated under the different supervision levels, allowing a direct comparison between unsupervised, par-

tially supervised, and fully supervised learning. This structure makes it possible to analyse how the learning strategy and the quantity of labelled data influence the final classification performance.

The results obtained from these experiments are used to compare the different training approaches and to evaluate the benefits of semi-supervised learning in both controlled and real-world scenarios.

4.1. Artificial Controlled Environment

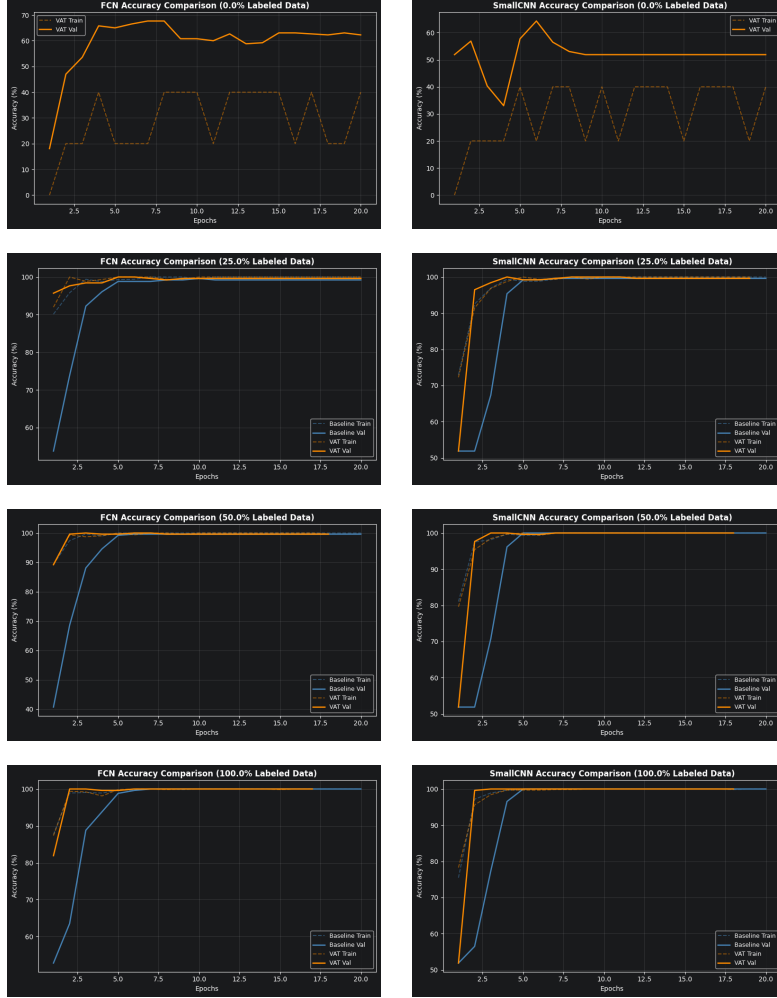


Fig. 2. Artificial Controlled Environment Results

Labeled Data (%)	FCN	vatFCN	baselineSmallCNN	vatSmallCNN
Unsupervised	66.19%	n/a	52.34%	n/a
25% Labels	98.44%	98.44%	98.83%	98.05%
50% Labels	95.77%	99.22%	99.61%	98.05%
Fully Supervised	96.95%	100.00%	99.61%	99.22%

Table 1. Classification Accuracy Comparison Across Label Availability

When no labelled samples were available (0% labelled data), only the VAT model could be trained. In this fully unsupervised setting, both architectures achieved validation accuracies above 50%, with the FCN reaching approxi-

mately 63–68% and the **SCNN** stabilizing around 52–64%. These results demonstrate that **VAT** is capable of learning meaningful decision boundaries even without direct supervision.

For the partially supervised scenarios (25% and 50% labelled data), both baseline and **VAT** models achieved very high performance. However, the **VAT** models generally converged faster and reached maximum validation accuracy in fewer epochs. The effect was particularly visible during the first training epochs, where **VAT** consistently produced higher validation accuracy than the baseline models. As training progressed, both approaches converged to almost perfect classification performance.

When all training samples were labelled (100% labelled data), the performance difference between the two training strategies became negligible. Both **FCN** and **SCNN** models rapidly reached validation accuracies close to 100%, indicating that the artificial dataset is relatively easy to separate when sufficient labelled data are available.

4.2. Real-World Experiments using the GTSRB Dataset



Fig. 3. Artificial Controlled Environment Results

When no labelled samples were available (0% labelled data), the **VAT** models were trained using only the unsupervised consistency objective. In this scenario, both architectures failed to learn meaningful traffic sign representations. The **FCN** achieved validation accuracies fluctuating between approximately 1% and 5%, while the **SCNN** remained

close to 1% throughout training. These values are substantially lower than the results obtained in the artificial experiments and indicate that completely unsupervised VAT is insufficient for solving a highly complex 43-class traffic sign recognition problem. The results suggest that some labelled information is necessary to guide the learning process when dealing with real-world image data.

For the partially supervised setting with 25% labelled data, both architectures showed substantial improvements. The FCN reached validation accuracies of approximately 76–77%, while the SCNN achieved nearly 100% validation accuracy. The influence of VAT was particularly visible during the early stages of training. Models trained with Virtual Adversarial Training consistently achieved higher training accuracy and converged faster than their baseline counterparts. The effect was especially noticeable for the SCNN, where the VAT model reached high validation accuracy after only a few epochs. These results demonstrate that unlabeled samples can provide valuable information when only a limited amount of labelled data is available.

When the amount of labelled data was increased to 50%, both architectures achieved higher performance. The FCN obtained validation accuracies around 82–84%, while the SCNN again approached perfect classification performance with validation accuracies close to 100%. Similar to the 25% scenario, the VAT models generally converged more rapidly and maintained slightly higher training accuracy during the learning process. However, the difference in final validation accuracy between the baseline and VAT approaches became smaller as more labelled data became available.

In the fully supervised setting (100% labelled data), the benefits of semi-supervised learning were further reduced. For the FCN, both the baseline and VAT models achieved similar validation accuracies between approximately 86% and 89%. The SCNN achieved near-perfect classification performance, reaching validation accuracies close to 100% regardless of the training strategy used. Although VAT still provided slightly faster convergence during the initial epochs, both approaches ultimately converged to nearly identical solutions.

4.3. Comparison with Established Deep Learning Classifiers

To provide additional context for the obtained results, several widely used image classification architectures were trained on the GTSRB dataset under fully supervised conditions. These benchmark models included YOLO26n-CLS [3], YOLO11n-CLS [4] and ResNet18 [2].

Model	Best Validation Accuracy
YOLO26n-CLS	86.2%
YOLO11n-CLS	86.1%
ResNet18	99.9%
FCN + VAT (100% labels)	89.0%
Small CNN + VAT (100% labels)	99.2%

Table 2. Comparison between benchmark architectures and the proposed models on GTSRB.

ResNet18 achieved the highest overall performance, reaching validation accuracies close to 100% after only a few epochs. The model exhibited stable convergence and minimal discrepancy between training and validation accuracy, indicating excellent generalization.

When compared with the proposed architectures, the fully supervised FCN achieved slightly higher accuracy than both YOLO classifiers, reaching approximately 89% validation accuracy. The SCNN substantially outperformed the YOLO models and achieved performance comparable to ResNet18, with validation accuracies approaching 100%.

These results indicate that although lightweight YOLO classification models provide competitive performance with relatively low computational complexity, deeper convolutional architectures remain more effective for traffic sign recognition on the GTSRB dataset. Furthermore, the proposed SCNN demonstrates that high classification accuracy can be achieved without relying on large-scale pre-trained architectures.

5. Conclusion

The objective of this research was to investigate whether a **VAT**-based semi-supervised learning (**SSL**) approach could be effectively implemented from scratch and applied to traffic sign recognition. To evaluate its effectiveness, two neural network architectures, a **FCN** and a **SCNN**, were trained and tested under unsupervised, semi-supervised and fully supervised learning scenarios using both artificial datasets and the **GTSRB** benchmark dataset.

The experimental results demonstrate that **VAT** can successfully exploit unlabelled data and improve the learning process when only a limited number of labelled samples are available. In the **artificial experiments**, **VAT** was capable of learning meaningful decision boundaries even without labelled data, achieving validation accuracies substantially above random guessing. Under partially supervised settings, the **VAT** models consistently converged faster and reached high classification performance using fewer labelled samples than the baseline models.

The **real-world experiments** on the **GTSRB** dataset revealed a more challenging learning scenario. While fully unsupervised **VAT** was unable to achieve useful classification performance on the complex 43-class traffic sign recognition task, the method proved effective once a modest amount of labelled data was introduced. In the 25% and 50% labelled settings, **VAT** generally accelerated convergence and achieved slightly better performance than conventional supervised training. As the amount of labelled data increased, the advantage of semi-supervised learning gradually diminished because the supervised objective already provided sufficient information for accurate classification.

A clear difference was observed between the two network architectures. The **SCNN** consistently outperformed the **FCN** across all experiments, achieving near-perfect classification accuracy on the **GTSRB** dataset in the partially and fully supervised settings. This confirms the importance of convolutional feature extraction for image recognition tasks, where spatial information plays a critical role. Although the **FCN** achieved competitive results, it was unable to match the performance of the convolutional architecture.

Overall, the results confirm that Virtual Adversarial Training is a practical and effective semi-supervised learning technique for traffic sign recognition when labelled data are scarce and unlabelled data are abundant. The study demonstrates that combining **VAT** with a convolutional neural network architecture provides the most effective solution, achieving high classification accuracy while reducing the dependency on costly manual data annotation. These findings highlight the potential of semi-supervised learning methods for real-world autonomous driving applications, where obtaining large quantities of labelled data remains a significant challenge.

6. Acknowledgement

This work is the result of our own activity, and we confirm we have neither given, nor received unauthorized assistance for this work. We declare that we used generative AI or automated tools in the creation of content or drafting of this document.

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Acronyms

AD Autonomous Driving. [1](#)

AI Artificial Intelligence. [1](#), [2](#)

DL Deep Learning. [1](#), [4](#), [5](#)

FCN Fully Connected Network. [1](#), [5](#), [6](#), [8–11](#)

GTSRB German Traffic Sign Recognition Bench-mark. [1](#), [5](#), [7](#), [10](#), [11](#)

KL Kullback-Leibler. [4](#)

LDS Local Distributional Smoothness. [4](#)

ML Machine Learning. [1](#), [2](#)

ReLU Rectified Linear Unit. [1](#), [6](#)

SCNN Small Convolutional Neural Network. [1](#), [5](#), [6](#), [9–11](#)

SL Supervised Learning. [1–5](#), [7](#)

SSL Semi-Supervised Learning. [1–5](#), [7](#), [11](#)

UL Unsupervised Learning. [1](#), [2](#), [7](#)

VAT Virtual Adversarial Training. [1](#), [4](#), [5](#), [7–11](#)